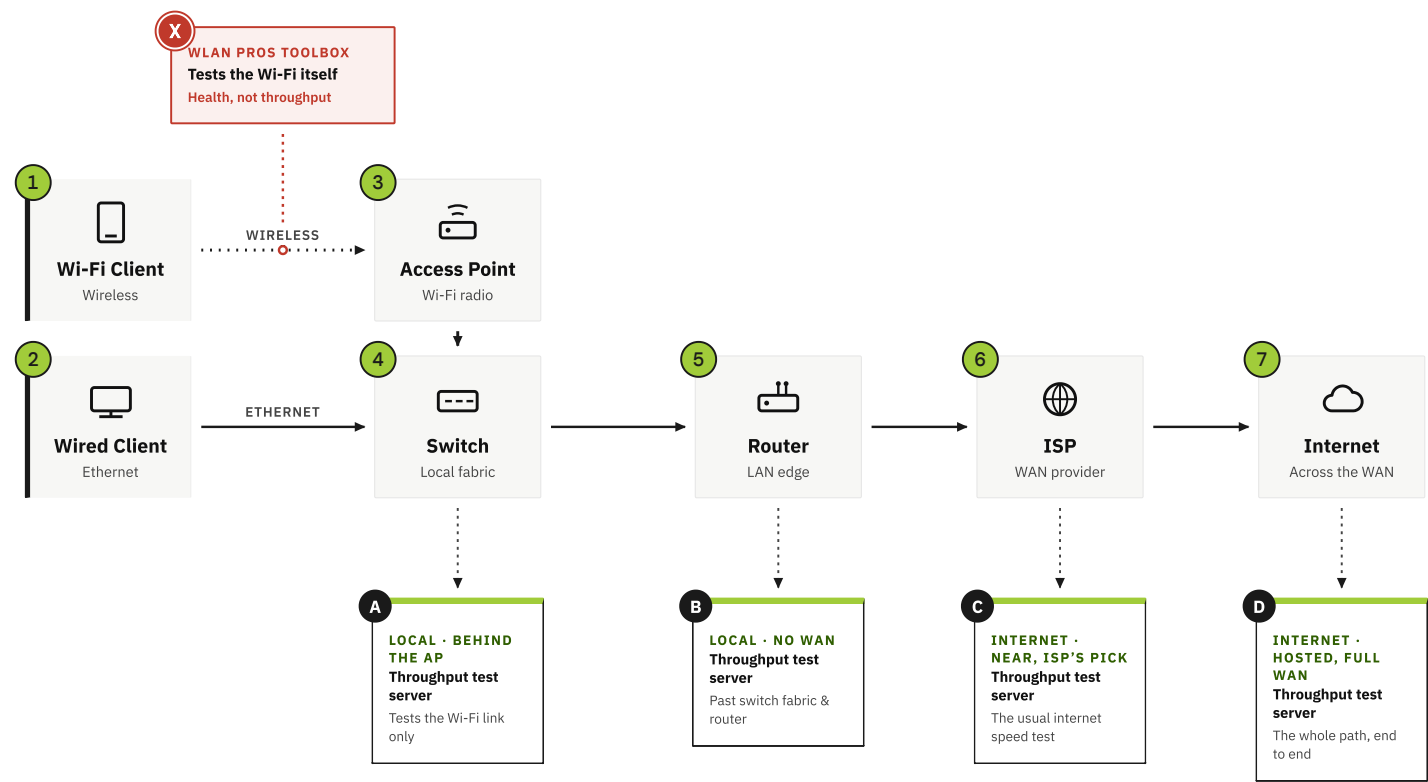


Where you test changes the number

A throughput test only measures the path **between the client you start from and the server you measure to**. The boxes **1–7** are the path; **A–D** are where a test server can sit along it; **X** is the Toolbox app testing the Wi-Fi itself. **What, where, when, and how** you test all change the result.



THE RESULTS MATRIX: WHAT EACH TEST ACTUALLY MEASURES

From 1 : Wi-Fi client From 2 : Wired client To X : Toolbox Wi-Fi test To A-D : test server

FROM	TO	WHAT IS BEING TESTED	RESULT
1	X	RF info, CH utilization, MCS rates, retry rates, data rates	Health of the RF & Wi-Fi network, without throughput
1	A	Wi-Fi to a test server directly behind the access point	Testing the Wi-Fi link only: reports Wi-Fi throughput
1	B	Wi-Fi through AP, switch fabric, and router, no WAN	Testing Wi-Fi plus local network, without WAN
1	C	Wi-Fi through local network to the server the ISP chooses	Testing local network plus WAN to the ISP's server
1	D	Wi-Fi & local network across the WAN to an internet-hosted server	Testing Wi-Fi plus local network plus WAN across the internet
2	A	Ethernet to a locally attached test server	Testing the local Ethernet only
2	B	Ethernet through the local switch and router, no WAN	Testing Ethernet plus local network, without WAN
2	C	Ethernet through local network to the ISP's chosen server	Testing local network plus WAN to the ISP's server
2	D	Wired local network across the WAN to an internet-hosted server	Testing local network plus WAN across the internet
1 2	D	Compare wired vs. wireless to the same endpoint server	Is it Wi-Fi, the local network, or the WAN that's the limit?

Before you trust a number, ask:

- ? When, where, and what client is being tested? Why?
- ? What is being tested: TCP, UDP, other? Why?
- ? Where is the test server located? Why?
- ? How often are tests completed? Why?
- ? One test, or multiple and averaged? What variability did you see?

LOCAL TEST SERVERS

Can be iPerf, LibreSpeed, a WLAN Pi, and more, and can be configured to mimic certain application loads. An internet speed test cannot.

Throughput testing: where you test changes the number. Boxes **1–7** are the path: a Wi-Fi client **[1]** over the air to the access point **[3]**, a wired client **[2]** into the switch **[4]**, then router **[5]** → ISP **[6]** → internet **[7]**. A server at **A** reports the Wi-Fi link; a server at **D** folds in the whole local network and the WAN. **X** is the Toolbox measuring RF and Wi-Fi health directly, with no throughput number at all. Same client, very different result.